

ABHAY KUMAR

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[LinkedIn](#) • [GitHub](#) • [LeetCode](#)

PROFESSIONAL SUMMARY

Full-Stack Software Developer at TCS with 1+ years of experience building and maintaining enterprise insurance applications using Java, Spring Boot, and Angular. I enjoy working across the stack — from writing optimised database queries and REST APIs on the backend, to fixing tricky UI bugs and improving component performance on the frontend. Outside of my core work, I have mentored two junior developers and led a client demo that brought in a major enterprise account.

WORK EXPERIENCE

Tata Consultancy Services (TCS) | TCS BaNCS — Health Insurance Platform

Software Developer | Java • Spring Boot • Angular • Microservices • PostgreSQL *July 2024 – Present*

- Engineered and maintained large-scale **Angular** modules for the TCS BaNCS health insurance platform, resolving critical UI bugs and optimising component rendering performance across **Policy Management** and **Claims Management** workflows, directly enhancing end-user experience and system reliability.
- Developed **RESTful APIs** using **Spring Boot** — covering CRUD operations, file upload/download, XML generation, and Excel processing. These APIs were used across multiple insurance modules to move data between services reliably.
- Deployed and maintained **Spring Boot microservices** as part of a distributed system, using REST-based inter-service communication and environment-specific configurations — ensuring the application ran consistently across dev, UAT, and production.
- Optimised backend data persistence using **Hibernate (JPA)** and **JDBC**, crafting 10+ performance-tuned **PostgreSQL** queries and applying composite indexing strategies — improving query execution speed by approximately **20–25%** and enhancing system responsiveness across high-volume data operations.
- Mentored 2 junior developers**, guiding them through onboarding, codebase familiarisation, and development best practices — accelerating their project readiness and strengthening overall team capability.

SKILLS

Programming Languages: Java, JavaScript, TypeScript

Frameworks & Libraries: Spring Boot, Spring MVC, Hibernate, Spring Data JPA, Angular, Next.js, Node.js

Databases: PostgreSQL, MySQL

Tools & Version Control: Git, GitHub, IntelliJ IDEA, Postman, JasperReports

Concepts: Microservices, RESTful APIs, Low Level Design, WebRTC, Data Structures & Algorithms

ACHIEVEMENTS

Star of the Month & On the Spot Award — January 2026, TCS

Recognised for exceptional contributions to a high-stakes client demo project that successfully attracted a major enterprise client, directly contributing to business growth for the organisation.

GATE 2024 Qualified — Computer Science & Information Technology

Cleared one of India's most competitive national engineering exams, demonstrating strong CS fundamentals and analytical ability beyond core ECE background.

AIR 1531 — TCS CodeVita Season 12 (2023–24)

Ranked AIR 1531 among 300,000+ global participants in one of the world's largest programming competitions, demonstrating strong analytical and competitive coding skills.

500+ Problems Solved — LeetCode & GeeksForGeeks

Consistent competitive programmer with 500+ problems solved across Data Structures and Algorithms, reflecting strong coding fundamentals and interview readiness.

PERSONAL PROJECTS

LocalDrop — Browser-based P2P File Transfer App | [Link](#)

Tech Stack: Next.js, TypeScript, WebRTC DataChannel, Socket.io, Node.js, Tailwind CSS — Deployed: Vercel (frontend) + Railway (signaling server)

- Built a file sharing app that transfers files directly between browsers using WebRTC DataChannel — no server ever touches the file, making transfers private and serverless by design.
- Diagnosed and fixed a production failure where SSE-based signaling broke on serverless infrastructure — replaced it with a persistent Socket.io server on Railway, which restored reliable peer connection across all environments.
- Tracked down corrupted file transfers to unordered DataChannel delivery — switched to ordered reliable transmission and added 256KB chunking with backpressure control to handle large files without memory overflow.
- Separated the system into two independent deployments — Next.js frontend on Vercel and a standalone Node.js signaling server on Railway — keeping concerns cleanly separated and each layer independently deployable.

AI-Powered Portfolio Website with Resume Chatbot | [Link](#)

Frontend: Next.js 15, React 18, TypeScript, Tailwind CSS, shadcn/ui | Backend: Python, FastAPI, Mistral AI, PyPDF2 — Deployed: Vercel + Render

- Designed and deployed a full-stack portfolio with a Next.js 15 TypeScript frontend on Vercel and a separate Python backend on Render — with dark/light theme toggle, WCAG accessibility compliance, and responsive layout across all devices.
- Developed a Python backend integrated with Mistral AI that fetches and parses the live resume PDF at runtime using PyPDF2 — so recruiters can ask questions about experience and skills and get accurate, context-aware answers through a multi-turn chat interface.

E-Commerce Web Application | [Link](#)

Tech Stack: React JS, JavaScript, Context API, JSON-Server, Tailwind CSS

- Engineered a fully functional e-Commerce application with user authentication, dynamic product filtering, and order management — dark-mode UI included for a polished end-user experience.
- Used React Context API for global state management across the app without prop drilling, and JSON-Server to simulate a RESTful backend for order persistence.

EDUCATION

Heritage Institute of Technology, Kolkata

2020 – 2024

B.Tech — Electronics and Communications Engineering | CGPA: 8.70 / 10

Kendriya Vidyalaya Kankarbagh

2019

Class XII — CBSE (All India Senior School Certificate Examination) | 84.8%